RAHEE

Project: RAHEE - a digital version of a travel board game
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1. What the project is

RAHEE is an existing board game by CCL, IIT Gandhinagar. Two to six players travel across a map of India. Your hand of playing cards is your itinerary: you play a card, move up to its value along connected routes, and pay ₹200 for every step out of a ₹25,000 budget. You score by visiting tourist places, reaching the destination goals printed on your character card, and collecting sets of themed memories.

We have to build a web version of the game that allows players to play online using multiple devices. Players can join the same game from their own devices and play together in real time. To make communication easier during the road trip, we will also provide both public and private chat options. The public chat will allow everyone in the game to discuss and share information, while the private chat will allow players to communicate individually with specific players. This will make the online road-trip experience more interactive, collaborative, and engaging.

The main requirement is not the game but how it is built. The engine must contain only the rules. The whole board cities, their positions, routes, route types, step counts, surcharges, tourist places, characters, and their destination goals must be stored in an editable data file that the program reads when it starts and draws over the supplied map image. Adding a city, changing a route or editing a character's goals must change the game without changing any code.

The rules themselves, taken from the official manual, are:

→ **Cards.** 7s, 8s and 9s are removed. A=1, 2-6 face value, 10=10, J=10, Q=K=5. Each player is dealt a fixed hand at the start, and nothing is ever drawn again, so the whole game is about 7 turns each.
→ **Movement.** Play a card, move at least 1 and at most its value. ₹200 per step, +₹4,000 for a flight, +₹1,000 for ferries.
→ **Memories.** Of the played cards' suit - ♠ Bazaar, ♥ Food, ♣ Language, ♦ Festival. Earned for moving exactly the card's full value, for ending where another player is, and for a road trip.
→ **Tourism.** 51 cities are numbered tourist places. End a move on one, and you may pay ₹2,000 to mark it. Once per player.
→ **Road trips.** End a move where other players are and invite them. Flip a participant's face-up card; everyone moves together on it. Companions pay you ₹100 per step; tourism costs ₹1,000 each instead of ₹2,000; and everyone gains a memory.
→ **Aces.** Play alone (1 step), as a +1 booster, or for one extra memory.
→ **End and scoring.** Ends when someone empties their hand or reaches 5 goals; whoever finishes first wins the round. 10 per tourist place, 5 per destination, 10 bonus for 5 destinations, 15 per same-suit triple, 20 per set of all four suits, −5 for every ₹500 over budget.

Scope: For now, our primary focus is on developing the web version so each player can play on their own device, allowing multiple players to connect and play together online in real time. The game will support 2-6 players, with the board generated from game data and all core game rules implemented. Players will also have public and private chat options to discuss the road trip with everyone or communicate individually with other players. We will also include save/resume functionality and a clear final score and winner summary. In a later phase, we will try to add AI bots so that if fewer human players are available, AI players can fill the remaining slots.

2. Input and Output:

2.1 Inputs:

The proposed system will require three main types of input.

1. Board Data:

The game will use a structured data file containing all the information required to represent the board and implement the game rules. This will include:

- Details of all cities and locations on the board.
- Position of each city on the digital map.
- Tourist locations and their corresponding attractions.
- Airports and other special locations.
- Connections between cities, including route type and movement requirements.
- Character information and their destination goals.
- Travel costs, surcharges, and other relevant prices.
- Card/deck configuration for different numbers of players.
- Scoring values and other game-specific rules.

Keeping this information in a separate data structure will make the game data-driven, allowing board information and game parameters to be modified without changing the core game logic.

2. Digital Board/Map:

The system will use a digital representation of the game's original board as the background for the interactive game. Cities, routes, tourist locations, player tokens, and other interactive elements will be positioned on top of this map using the corresponding board data.

3. Player Actions and Game Setup:

At the beginning of the game, players will provide information such as:

- Number of players.
- Player names.
- Order of play.
- Character selection for each player.

During gameplay, the system will receive player actions such as selecting and playing cards, moving between connected cities, using special cards or actions, visiting tourist locations, participating in road trips, and ending a turn. For hotseat gameplay, players will also confirm the handover before the next player's turn so that private information remains hidden.

2.2 Outputs:

Based on the provided data and player actions, the system will generate the following outputs:

- **Interactive Game Board:** Displays the map, cities, routes, player tokens, available moves, and relevant destinations or locations.
- **Player Information Panel:** Displays the player's available budget, spending, cards, collected memories, visited tourist locations, and character-specific destinations while keeping private information hidden from other players.
- **Game Status:** Shows the current player, available actions, movement options, and important game events.
- **Final Scoreboard:** Displays each player's score with a clear breakdown of scoring components, making the final result easy to understand and verify.
- **Save and Resume:** The system will attempt to save the current game state so players can continue an unfinished game later.

3. Methodology:

3.1 System Architecture: The proposed web version will follow a modular and data-driven architecture. The game board, cities, routes, costs, and other game-specific information will be stored separately from the game logic. This separation will allow the game data to be modified or updated without requiring major changes to the underlying code.

The system will consist of four main components:

Board Data → Game Engine → User Interface → Main Application

The Game Engine will be responsible for managing the rules and current state of the game. It will handle player actions, card dealing, movement, costs, memories, tourism, road trips, Aces, end-game conditions, and scoring. The engine will remain independent of the user interface so that the game logic can be tested separately.

The Board View will display the game board, routes, cities, player tokens, and relevant highlights. The Sidebar will display information such as memories, visited places, and destinations. The Screens component will handle game setup, player joining, private handover screens, and the final scoreboard. The Main Application will coordinate navigation between screens and player interactions.

3.2 Data-Driven Game Board: The game board will be created from structured game data rather than being hard-coded into the application. The data will contain information such as cities, connections between cities, route types, movement costs, and other properties required by the game rules.

This approach will make the system flexible and easier to maintain. If a city, route, cost, or other game property needs to be changed, the corresponding data can be updated without rewriting the core game logic. It will also allow the same game engine to work with different board configurations in the future.

3.3 Game Logic and Algorithms: The game will represent the board as an undirected graph, where cities are represented as nodes and routes between cities are represented as edges. The graph will be generated from the board data when the game starts.

Standard algorithms will be used wherever required:

- **Graph representation** will be used to model connections between cities.
- **Neighbour traversal** will be used to determine legal moves based on the player's available movement.
- **Breadth-First Search (BFS)** will be used to identify locations reachable within a specified number of steps.
- **Dijkstra's algorithm** may be used later to provide route hints as a stretch feature.

- **Set-based scoring logic** will be used to calculate memory-related scores in accordance with the game rules.

No new AI or pathfinding algorithm will be developed as part of the core project. The focus will be on correctly implementing the existing game rules and providing a reliable digital version of the physical game.

3.4 Gameplay Flow: The basic gameplay flow will be:

Game Setup → Players Join → Private Handover / Player Turn → Select Card → Move Player → Resolve Actions → End Turn → Next Player → Final Round → Final Scoreboard

For online multiplayer, each player can connect from their own device and participate in the same game in real time. The system will maintain a synchronized game state so that all players see the relevant updates as the game progresses.

A private handover screen will also be provided for single-device or hotseat gameplay. This will allow players to pass the device between turns while keeping private information, such as cards in hand, hidden from other players.

3.5 Multiplayer and Communication: The initial focus of the project will be to enable players to play together online using their individual devices. A player can create or join a game, and multiple players can connect to the same game session.

To support communication during the road trip, the application will provide both public and private chat. The public chat will allow all players in the game to discuss the trip and gameplay together, while the private chat will allow a player to communicate directly with another player.

3.6 Save and Resume: We will try to include save and resume functionality, allowing players to save their current game state and continue the game at a later time. The system will also aim to provide a clear final scoreboard showing each player's score and the overall winner.

3.7 Testing Strategy

The application will be tested at both the game logic and user-interface levels.

1. **Game Logic Testing:** Test individual rules, including card dealing, movement, costs, player actions, tourism, road trips, scoring, and end-game conditions.
2. **Complete Game Testing:** Run complete games with different player counts to verify that the game progresses correctly from setup to the final scoreboard.
3. **Rule-Based Test Cases:** Use examples and scenarios provided in the original game rules to verify that the digital implementation produces the expected results.
4. **Multiplayer Testing:** Test multiple players joining from different devices and verify that game actions, turns, and state changes are correctly synchronized.
5. **Interface Testing:** Test the application on different screen sizes and devices to ensure that the game remains usable and understandable for all players.

4. Assumptions to confirm:

The manual is silent on these, so we chose an answer and wrote it down. The first two affect the design most.
1. The Pilgrim's goal printed as "34 Agartala" is read as Agartala (33), since the stamp list has 33 = Agartala and 34 = Kolkata.
2. On meeting a traveller, only the moving player gains the memory.
3. You may not pass - you must play a card while you still have one.

References:
1. Rahee — A Traveller's Journey Across India, official manual. Design: Jyothi Krishnan & CCL, IITGN. Art: Parvathy Subramanian.